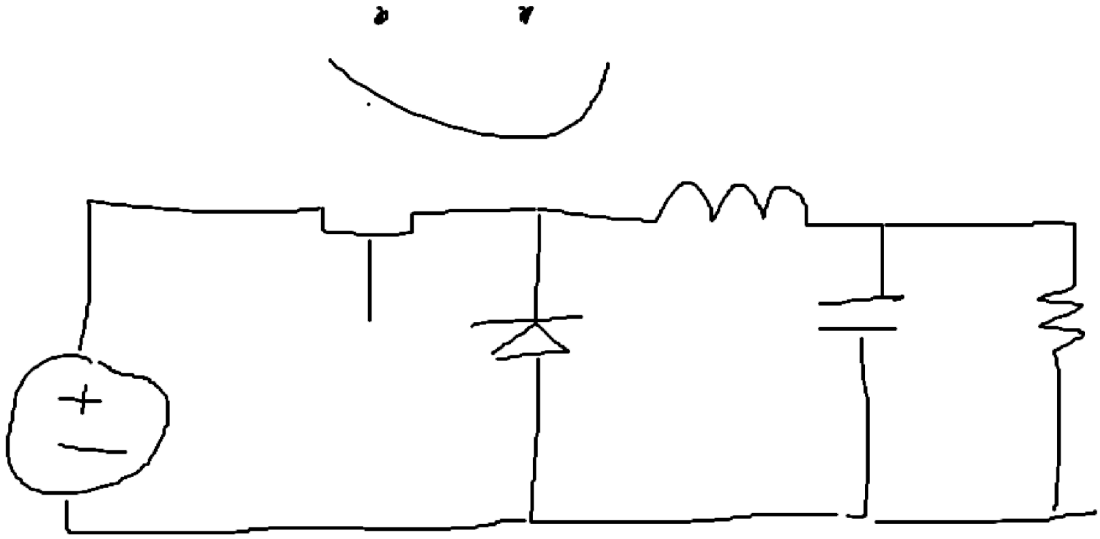
Formula Slug – Buck Converter Schematic Block
Power Electronics Onboarding Project 2026-2027

Overview

This project is centered around designing a buck converter to help power different boards on our LV system. This project requires you to design your own buck converter, sourcing components and creating the schematic and layout. You will also push it onto FS's github.



Our Low Voltage system consists of all our important logic and sensing boards, such as our Vehicle Control Unit, Telemetry boards, and Tractive Battery board. On all these boards, there are a variety of different components, who all may need to be powered by a different voltage level. For example, relays can operate at 12V, MCUs can operate at 5V, while some smaller chips run on as small as 3.3V. This is a big issue, because our low voltage bus runs on 12V!? If we overvolted those components, they would immediately stop working.

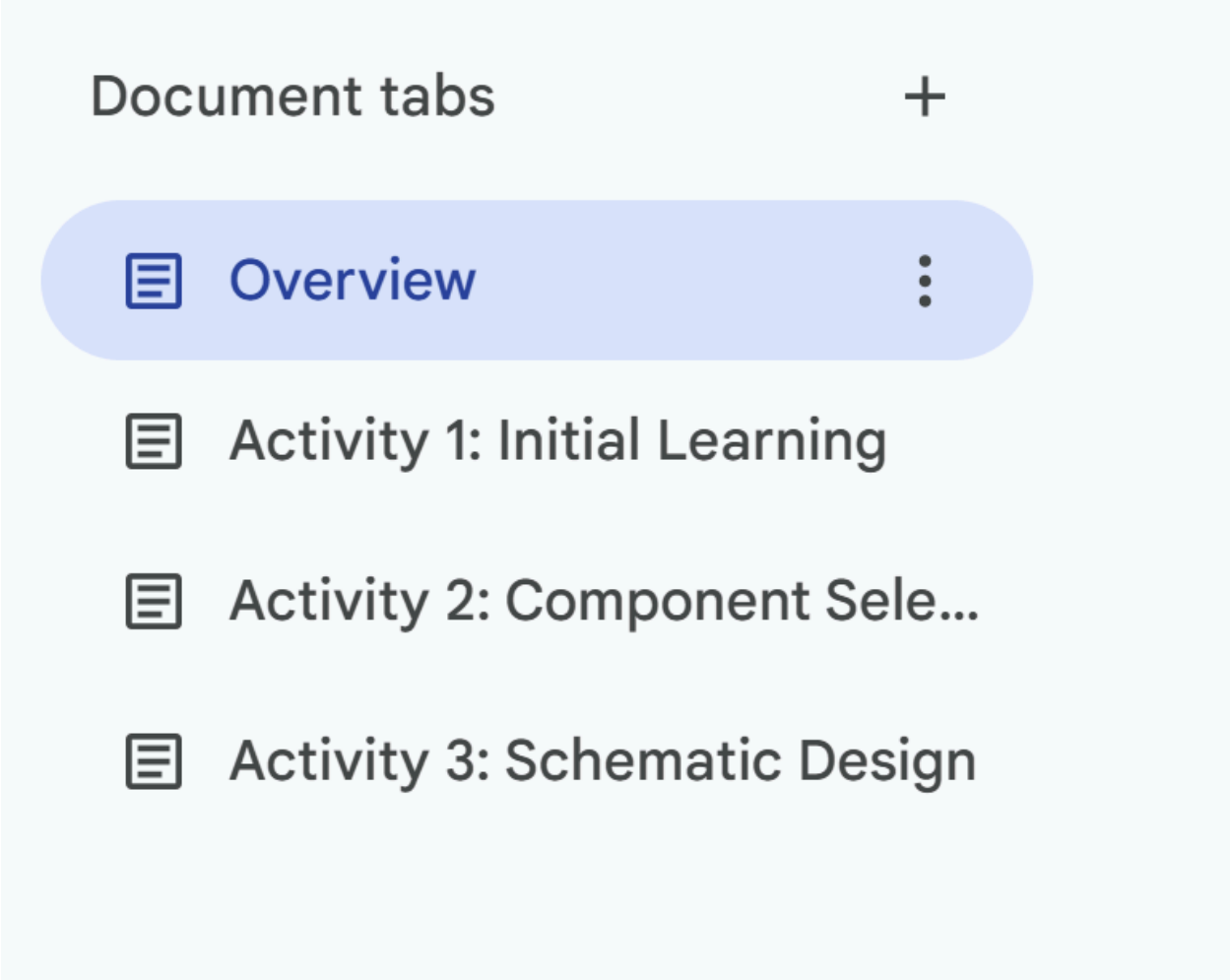
So what can we do to fix this?

A buck converter is used to transform a higher voltage to a lower voltage. We can use this to

power all of these components that require a much smaller power input, allowing it to properly function.

Navigating Project & Submission

There are 3 parts to this project: Initial Learning, Component Selection, and Schematic Layout. On the left side of the Google Doc, you will find the different tabs leading you to each activity.



Make a copy of this project, renaming it to "*Full Name* - Power Electronics Onboarding". Once every activity is completed, or before the due date, submit the document here. For example:

"Edmund Lai - Power Electronics Onboarding"

If there are any questions, message Edmund Lai on slack. There will be updates to this document if needed, check below for update logs.

Revision	Date	Updated Content
1.0	9/14/2026	Initial draft

Activity #1: Initial Learning

NOTES WRITTEN IN RED

Listed below is a variety of initial resources that you can use if you are not familiar with electrical engineering concepts. I would suggest getting to know these well before you attempt to work on the buck converter. Any resource that talks about the same thing is good, as long as you understand the basics.

Analog Devices and Circuits:

1. Organic Chemistry Tutor: Ohm's Law: https://www.youtube.com/watch?v=_rSHqvjDksg
    - V = IR, Voltage(V) = Volts, Current(I) = Amps, Resistance(R)= ohms
    - V up, I goes up(directly proportional).. R goes up, I goes down(inversely proportional)
    - Current flows from positive to negative
    - SERIES CIRCUIT:
        - EX:
        - CURRENT going through is the same
    - Kirchhoff's Voltage Law: As you go through a circuit in a closed loop, total voltage is 0, as the battery increases circuit energy, and resistors consume energy.
    - Kirchhoff's Current Law: Current that enters the junction = current that leaves the junction
    - PARALLEL CIRCUIT:
        - EX:
        - EX KIRCHHOFF'S CURRENT LAW:
        - VOLTAGE connected across resistors in parallel is the same

2. Khan Academy, Current, Resistors, Ohm's Law: https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/circuits-part-1
    - Voltage = Electric Potential Energy per unit charge (Volts)
    - Current = Charge per Unit time; how much charge flows past a point in a circuit in a second
    - Resistance: How much charge is impeded ohm
    - Convention for I, positive to negative terminal
3. Khan Academy, Compound Circuits: https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/electric-power

    Why do Resistors heat up?
    - Current passes through a resistor, and Voltage changes, meaning electric potential energy changes

- The loss in electric potential energy gets transferred into the atoms and molecules that are passed through, leaving a resistor that's hotter
- DERIVATION OF V = IR: To calculate the amount the resistor will heat up by: user P = W/t =ΔE/t=ΔUe/t
    - ΔUe = qV
      qV2 - qV1 = ΔUe
      P = q(v2-v1)/t ..(q/t is current)
      P = IΔV = J/s = Watts OR P = I^2R OR P = (ΔV)^2 / R

4. EEVBlog, Voltage Dividers: https://www.youtube.com/watch?v=xSRe_4TQbuo&list=PLvOlSehNtuHtWlH0UOZNtOn-FlFCn1GYw&index=2
    - Voltage Divider(RESISTORS IN SERIES): A circuit that uses resistors to create a smaller voltage from a larger one
        - Commonly used for:
            - Voltage Sensing = reducing a voltage to a level an electronic chip can safely measure
            - Feedback circuits = help the controller maintain the correct voltage
            - Reference = a known voltage used as a comparison point; bias voltage = a voltage used to establish how a component operates
            - Potentiometers = adjustable resistive divider that can adjust resistance between top and bottom
        - Formula: Voltage Divider:
            - Vout = Vin((R2)/(R1 + R2))
            - Vin = input voltage, Vout = Output voltage
            - R1 = top resistor, R2 = bottom resistor
            - Increase the top resistor (R1); the output decreases. Increase the bottom resistor(R2); output increases
    - Resistance Measurement:
        - Series(Current(I) is the same): Rtotal = R1 + R2
        - Parallel(Voltage(V) is the same): 1/Rtotal = 1/R1 + 1/R2 OR SHORTCUT FOR TWO RESISTORS: (R1R2)/(R1 +R2)
    - Current Dividers(RESISTORS IN PARALLEL): A circuit that uses parallel resistors to split an incoming current into smaller currents for different branches
        - Formula:
            - I1 = Itotal((R2)/(R1+R2))
            - Numerator: the Other resistor; R1 is the resistance of that circuit
    - Thévenin's and Norton's theorems
        - Thévenin's theorem(Restistor IN SERIES) = a complicated linear circuit can be represented by 1 voltage source and 1 series resistor
            - Vth = Thévenin's voltage = open-circuit voltage at output terminals

        - Rth = Thévenin's resistance = Equivalent resistance looking into terminals w/ independent sources turned off
    - Norton's theorem(Resistor in Parallel) = the same circuit can be represented by 1 current source and 1 parallel resistor
        - In = Norton current = short-circuit current at output terminals
        - Rn = Norton Resistance = equivalent resistance seen at terminals

- Circuit Loading:
    - Device is connected to a circuit's output and draws current; it can change the output voltage
    - Voltage divider; load connected between the output and ground acts parallel with bottom resistor, causing it to drop in voltage
        - Theres an extra path for the current to flow through, and that path changing how much current flows to the top resistor
        -

5. EEVBlog, Kirchhoff's Laws: https://www.youtube.com/watch?v=WBfAEeEzDlg&list=PLvOlSehNtuHtWlH0UOZNtOn-FlFCn1GYw&index=13

- KCL(Kirchhoff Current Law) = Sum of Current In = sum of Current Out
    - +Sum of Current in - Sum of current out = 0
- KVL(Kirchhoff Voltage Law) = Sum of Applied V around Loop = Sum of Voltage Drops around loop

Listed below are more power related resources. This you will need to use to understand the theory behind how switch mode power supplies work.

1. MIT OpenCourseWave Power Electronics Course:
    a. Lecture 5: Intro to DC/DC, Part 1

- DC(constant voltage)/DC Converter = An electrical circuit that changes one DC voltage level into another DC voltage level
    - **Buck converter: Steps voltage DOWN (12 V → 5 V).**
    - Boost converter: Steps voltage UP (5 V → 12 V).
    - Buck-boost converter: Can step voltage up or down
- How does Buck Converter lower voltage?
    - **Buck Converter IC: An integrated circuit that controls the rapid ON/OFF switching in a buck converter. It uses feedback to adjust switching and maintain the desired output voltage**
    - Inductor and capacitor smooth everything so output stays near a certain V
        - Inductor(L) smooths current: stores energy in a magnetic field and resists sudden changes in current to keep current flowing

        - Capacitor(C) smooths voltage: stores energy in an electric field and supply/absorb current to reduce fluctuations in output voltage
- Duty Cycle: percentage of each switching cycle that the main switch stays on
    - **For IDEAL BUCK CONVERTER: Vout = D(Vin)**
- Switching Frequency:
    - How many times the converter completes an ON/OFF cycle every second
    - **For assignment: 500 kHz(500,000 CYCLES PER SECOND)**
- Output Voltage Ripple: Small fluctuations in converters
    - **1 - 5% Ripple for project: 1% of 5V = 0.05V OFF, 5% of 5v = 0.25 V OFF**
- **Feedback controller:**
    - How the converter maintains the V when/if the input changes
    - 1. Measures the output voltage(through a resistor divider), 2. compares it to the reference voltage, 3. and adjusts the switching duty cycle to bring the output back towards 5V
- Average Values:
    - Periodic Steady state: converter repeats the same voltage and current patterns every switching cycle
    - Inductor volt-second balance: Over a complete repeating cycle, average voltage across inductor is 0
    - Capacitor charge balance: over a complete repeating cycle, average current through capacitor is 0

2. Fundamentals of Power Electronics, by Robert W. Erickson and Dragan Maksimovic
3. Power Electronics by Daniel W. Hart

Please take good notes! Take notes in your notebook, here, whatever you would want to do. I would just like to see what you guys see and how you understand it!

Activity #2: Component Selection

EXPLANATION FOR RESEARCH AND SELECTION OF EACH COMPONENT WRITTEN IN RED AT THE END.

After finishing Activity #1, you are ready to start selecting different components! When selecting components, please follow the LV Standards guideline to picking different form factors. Typically, our resistors and capacitors are both 0603, but this is just a general guideline. Typically do not go under 0603, but if we need a larger voltage rating, going up is completely fine. We typically only source from Digikey as well (Mouser if it is the only place available). We also do not want to use marketplace products. These are all filters that you can apply on Digikey, and more information is available in the LV standards document.



The first big thing to source is your Buck IC! This chip is used to adjust the duty cycle of the switches, and work with the feedback loop. For reference, here is the one we used last year but is now out of stock. The big thing is making sure it can output enough current to supply our boards! For FS5, we defaulted to a 2A buck, which worked perfectly fine (to my knowledge), so anything 2A and above is good! However, be sure to properly select this, because there are some tradeoffs that you may have to make if you oversize the buck converter too much. Feel free to ask me if you have any questions about that!

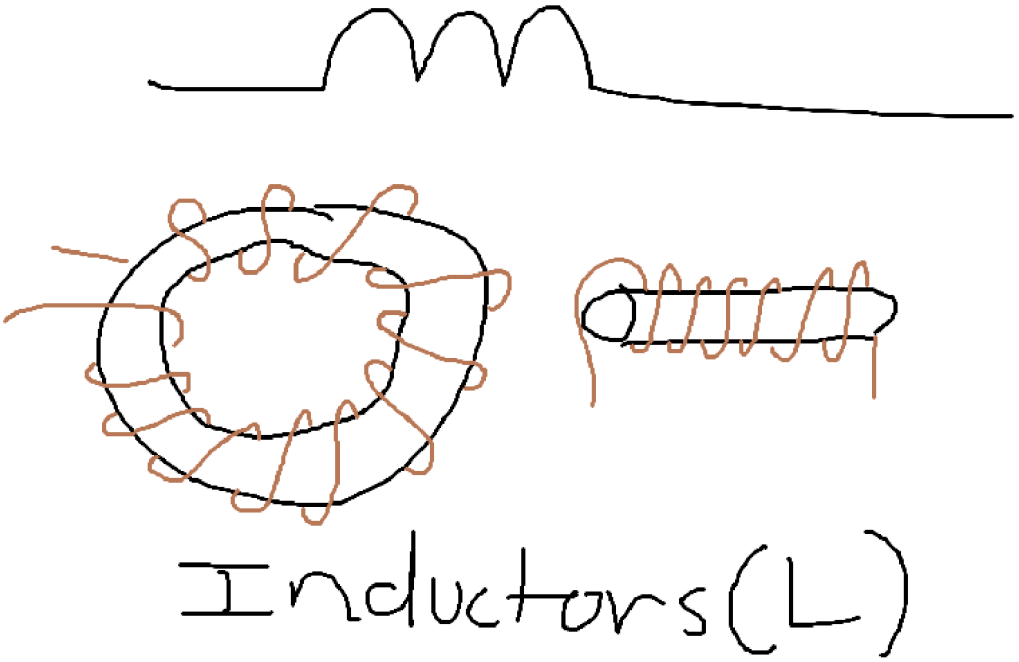
There are a couple I have in mind, but I would like you guys to do your own research and play with Digikey to be more familiar with the website! This is where we order almost all our parts,

excluding some from Mouser. You will need to get familiar with it. Here are some initial parameters to get you all started on looking.

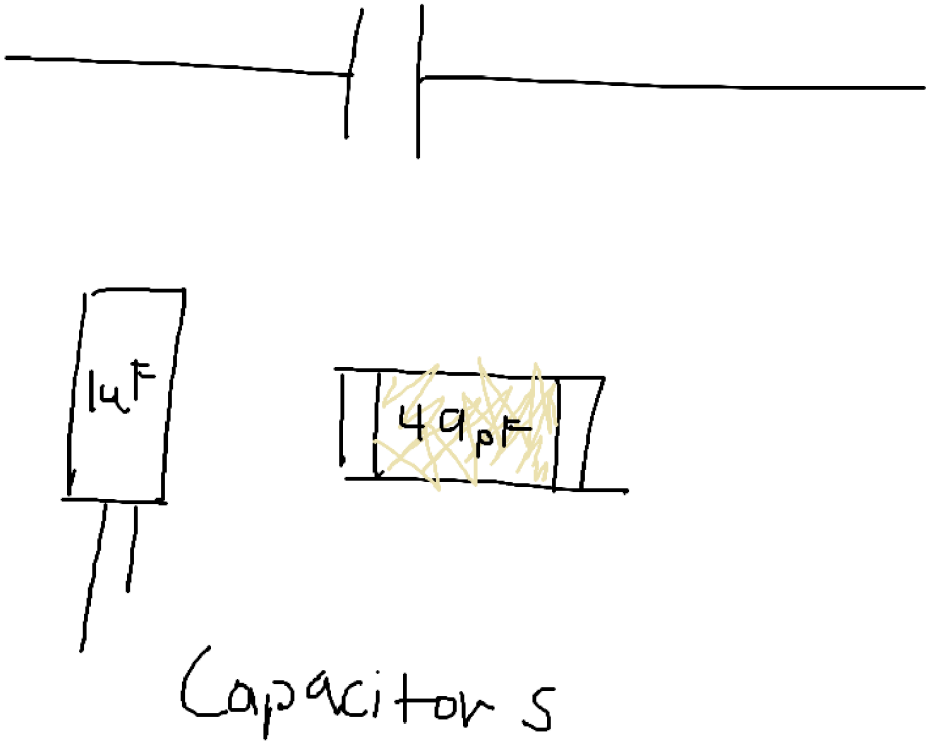
Input Voltage	~10-15V *
Output Voltage	5V
Output Current	Iout ≥ 2A
Switching Frequency	Fsw ≥ 500kHz**
Feedback Control	😳
Ripple Voltage	1-5%

*Though our LV bus is 12V nominally, it is actually 13.5V!
**with too low of a switching frequency, you upsize your inductor and output capacitor value, which could be an issue space wise. This is an arbitrary value I have set for the case of the onboarding project.

Whatever chip you decide to pick, be sure to properly read the entire thing, understanding what each pin does, what kind of internal structure it has, and the reasoning behind all of it.



One big thing you should check in an inductor is its saturation current. Make sure that your inductor does not oversaturate at the output current you are expecting, or else your inductor's inductance will drop so low that your output ripple will be extremely large.



Capacitors have a funny little thing called DC Biasing. Capacitors do not like staying at the specific capacitance they were designed to be at. Instead, whenever you apply any kind of voltage across one, their capacitance will drop by some percentage, typically shown in your datasheet. Make sure that your capacitances will be what you are expecting at your expected output voltage! A little hint: The output capacitors are there to help filter noise! There is something you have to do when picking your output capacitors to help reduce noise on the output end.



Make sure to try to minimize costs as much as possible! This is more of a good engineering practice to not just go with the best thing if it costs an unreasonable amount of money.

At the end of this activity, fill out this chart with the things that you have selected for ONE buck converter, with the costs per unit and size as well. If you need to add more rows, feel free to.

Component	Cost	Link	Units	Size/Package
Buck IC	$0.44	https://www.digikey.com/en/products/detail/texas-instruments/TPS563208DDCR/5825114?s=N4IgTCBcDaICoFMAeBDAzgAgJIDs0BcAnAVwFsEd9M4AFAZQFYA2AZjAAYAOAEW4GEASiAC6AXyA	1	SOT-23-6
3.3 µH Inductor	$2.51	https://www.digikey.com/en/products/detail/w%C3%BCrth-elektronik/744778003/9859314?s=N4IgTCBcDaIOoB%2BBOAXAFgAgKIBsCmA1ikgPYB2AlgRgOwAsdNNAHAAysDMIAugL5A	1	7.3 x 7.3 x 3.4mm
22 µF, 16V Input Capacitor	$1.26	https://www.digikey.com/en/products/detail/taiyo-yuden/MSASG32MSB5226MPNA01/16658591?s=N4IgTCBcDaICoEMCWBPA9gAgJoFcAmApgHYYCyAygILkDiAzGBQEICsYYAbKQAoBylABgCMIALoBfIA	2	1210
22 µF, 16 V Output Capacitor	$0.55	https://www.digikey.com/en/products/detail/tdk/C3225X5R1C226M250AA/569062?s=N4IgTCBcDaICoBEDSACAwgZjGArADRwCUBGNbANgFlcAGAQTpAF0BfIA	2	1210
0.1 µF, 50V Bootstrap Capacitor	$0.14	https://www.digikey.com/en/products/detail/kemet/C0603C104K5RACTU/1465594	1	0603
54.9 kΩ, 1% Feedback Resistor	$0.10	https://www.digikey.com/en/products/detail/panasonic-industry/ERJ-3EKF5492V/196403?s=N4IgTCBcDaIAoEMB2CDOB7JBLAxgAgFEAlAKQFoBmAgaQDEBWAFgE4wA1EAXQF8g	1	0603
10 kΩ, 1% Feedback resistor	$0.10	https://www.digikey.com/en/products/detail/panasonic-industry/ERJ-3EKF1002V/196066?s=N4IgTCBcDaIAoEMB2CDOB7JBLAxgAgFEAlAKQFoBmAgaQDEBGABk	1	0603

Component	Cost	Link	Units	Size/Package
		bADUQBdAXyA		
Total Cost	$6.91			

BUCK CONVERTER IC: TPS563208DDCR

The IC controls the switch that converts the higher input voltage to a lower regulated 5v output voltage. I selected the Texas Instruments TPS563208DDCR synchronous buck converter because it meets the main voltage(step down from 10/15V to 5V) and current requirements(supplying at least 2A) for the project. This IC is rated up to 3A, which also provides some current margin. The typical switching frequency is 580 kHz, which is above the project's 500 kHz minimum requirement, and I verified that it stays near 600 kHz across the 10-15V input range. This IC also integrates the switching MOSFET, so I didn't need to buy more external components. The small SOT-23-6 package also saves PCB space, but that also risks overheating if the traces are not placed and routed carefully.

INDUCTOR: 744778003

The inductor stores energy during switching and smooths the current. I chose a 3.3 µH Würth Elektronik inductor because of the buck converter's switching frequency, output current, and inductor saturation current. The stated saturation current rating is 4.1 A, which is higher than the peak current of around 2.87A(at a 15V input). This satisfies both the current requirement of atleast 2A output current and helps reduce the risk of saturation and excessive current ripple. I also considered the inductor's size and cost to make sure it was practical.

INPUT CAPACITORS(C1 AND C2): MSASG32MSB5226MPNA01

The input capacitors stabilize the input supply and provide current during switching. I chose two 22 µF input capacitors because they have a 35v rating that provides a voltage margin above the maximum input voltage(10V/1V). The onboarding document also emphasized DC biasing for ceramic capacitors. Because their effective capacitance can decrease when voltage is applied, I decided to use two capacitors in parallel to provide more input filtering. I used the 1210 larger footprint size rather than 0603, as the higher capacitance needs larger components, and the LV standards allow me to use them when necessary.

OUTPUT CAPACITORS: C3225X5R1C226M250AA

The output capacitors store charge and stabilize the output voltage during switching. I chose two 22 µF output capacitors based on the requirement of 5V output and 1-5 percent voltage ripple. I chose a 16V voltage rating because it is above the 5V output. Again, because ceramic capacitors can lose capacitance, I used two capacitors in parallel.

BOOTSTRAP CAPACITOR: C0603C104K5RACTU

The bootstrap capacitor provides the voltage needed to turn on the high-side MOSFET during switching. I chose a 100 nF capacitor based on the requirements from TPS563208DDCR(I used these requirements for the other parts too). I used a 50 V-rated capacitor to provide voltage margin and a 0603 footprint based on LV component guidelines.

FEEDBACK RESISTORS: ERJ-3EKF1002V

Feedback resistors create a voltage divider that sends a portion of the output back to the IC. I again followed the TPS563208DDCR requirements and chose R1 = 54.9kΩ and R2 = 10kΩ for a 5V output. I also chose a 0603 footprint based on LV component guidelines.

INPUT AND OUTPUT CONNECTORS:

Input and output connectors bring power to the circuit and connect the output to an external load. I didn't realize I needed them until I started building my schematic in KiCad, so I have not selected a specific connector part, and it is not included in my final cost. In KiCad, I added two-pin connectors, J1 and J2, for the input and output. I used a 2.54 mm pin header. I would need to finalize these choices before manufacturing the PCB.

Activity #3: Schematic and Layout Design

Here is where you start designing the schematic! Create a branch in the onboarding repo, and use the template KiCad project to make your buck converter. Be sure to follow the LV standards guide.

Please attend one of the KiCad workshops, where the electrical leads will teach you more about the software and what we expect.

Once you feel that your schematic and or layout is done, make a commit on the fs-electrical-onboarding repository, and push it onto github. You should make a commit no matter what, even if you did not finish completely. If you do not, you will not be considered for Power Electronics.

1. MAKING SCHEMATIC:

In KiCad, I connected the buck converter IC to the inductor, input and output capacitors, bootstrap capacitor, and feedback resistors based on the IC datasheet. Through this, I learned how the different components are electrically connected, how to assign footprints to symbols, represent them in the PCB editor, and, in general, how to turn a circuit diagram into an actual PCB layout.

2. ASSIGNING FOOTPRINTS:

After the schematic, I assigned the PCB footprints to the components to represent the physical size and pads where its soldered on the PCB.

3. PCB LAYOUT:

I transferred my schematic into PCB Editor and arranged the components onto the board. I then routed and rerouted copper traces to connect the components as I did in the schematic. I spent a lot of time rotating, arranging, and retracing the components to make the routing possible. Fixing this issue and figuring out my wire placement made me realize how much component placement and trace routing affect the electrical noise, heat, and how well the circuit performs.

4. ELECTRICAL RULE CHECK:

I ran the KiCad Electrical Rule Checks and kept checking for possible connection problems. My final ERC shows 0 violations, with 4 ignored tests. I learned how to fix the errors presented in the ERC by adjusting my schematic's electrical connections.

5: DESIGN RULE CHECK:

After completing the PCB layout, I ran the Rule Check and encountered several warnings(around 25). One issue involved the silkscreen near the buck converter IC. I corrected these issues by rerouting copper traces, rotating components, and adjusting the silkscreen converter IC around the buck converter IC so they didn't cause warnings. I kept running the DRC until I finally had 0 violations. This process taught me how to use DRC to find and fix the

PCB layout problems, but I also learned that passing DRC doesn’t guarantee the functioning of the board, as heat and electrical performance still need to be considered.

Extra Activity: Tractive System Shunt Current and/or Voltage Sense

If you happen to finish the buck converter, or you feel that the buck converter doesn't speak to you, you can also work on this project as well. This one will be in the form of a presentation / research project.

This project is based around adding to our current sensors within our Tractive Battery's container. Currently, we use a hall effect current sensor, which kind of looks like this:



For accuracy and data acquisition reasons, it would be nice to be able to add to this hall effect sensor with a shunt one, where you measure the voltage drop across a specially manufactured piece of metal with a specific resistance and voltage drop across, where you can then determine what current is going through it based on it. However, there are some constraints to this system, ie: isolation, heat dissipation, etc

Our current tractive battery current can go up to ~600A peak. We want to be able to measure up to this and down to the negatives as well in order to track regen current. Required by FSAE 26-27 Rules (EV.8.7.4), we need to be able to *prove* to judges that the Brake System Plausibility Device works through simulating tractive system current going through the current sensor. Find a way to be able to simulate this using a shunt if possible. At the moment, we have wire wrapped around a hall effect sensor multiple times, simulating higher currents with a smaller test current.

We would also like to be able to properly measure the voltage in our tractive system while we drive. We should find a way to take that voltage measurement, isolate it and pass it through to a

microcontroller so that we can have this data for the future.

When completing this research project, continue to write good documentation and make sure to optimize for costs when possible.

Have fun!